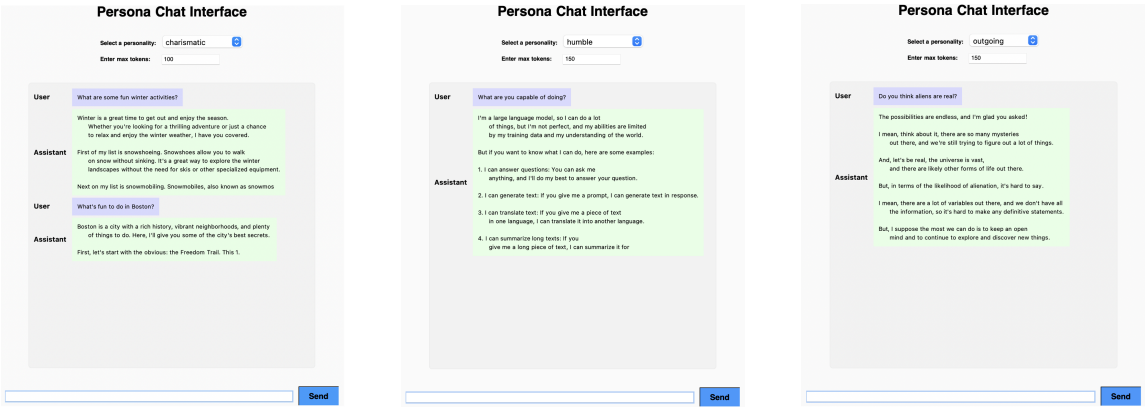

(a) Charismatic (b) Humble (c) Outgoing

Figure 10: Chat interfaces for interacting with different base personalities: (a) Charismatic, (b) Humble, and (c) Outgoing.

principal component weights (Figure 11). The interface also visualizes the designed personality and highlights its similarities to existing personality profiles in our lexicon. Figure 12 shows the chat interface for interacting with a custom-designed personality.

Due to potential safety and ethical considerations, we have chosen not to make this interface publicly available at this time.

4 Potential Use Cases

The ability to dynamically alter personality features in Large Language Models (LLMs) without substantial retraining opens up a wide range of creative uses. In the entertainment industry, this technology has the potential to alter interactive storytelling: envision video game characters modifying their personalities in real time, resulting in profoundly immersive and responsive narratives. Concurrently, in the field of personal technology, AI companions could fine-tune their qualities to match users' tastes or emotional states, resulting in more meaningful human-AI interactions.

Beyond entertainment, this ability has potential for improving professional services and education. Customer support systems could customize their communication styles for each client, potentially increasing customer satisfaction. In academia, AI-driven tutoring platforms may tailor their teaching avatars to students' learning preferences, potentially increasing engagement and knowledge retention. However, as we pursue these fascinating prospects, we must be cautious about the ethical implications and potential societal consequences of such personality-adaptive AI systems.

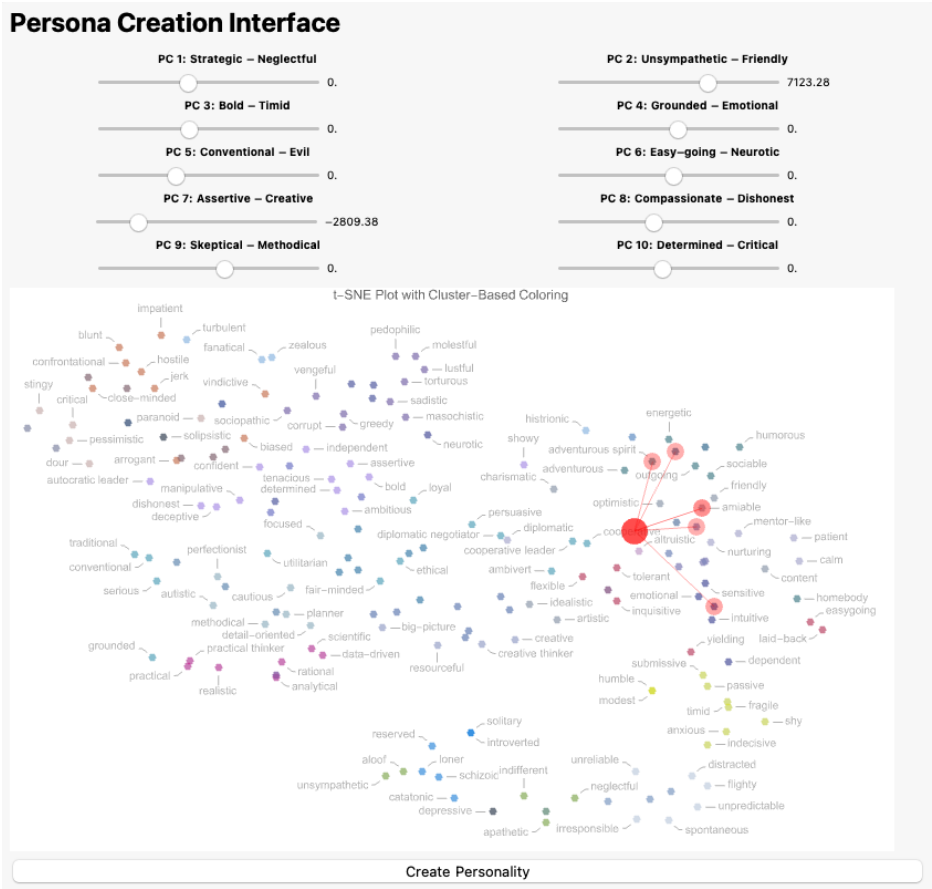

Figure 11: Interface for designing custom personalities through principal component manipulation.

5 Ethical Considerations

While the ability to dynamically adjust personality traits in LLMs offers a range of exciting applications, it also raises significant ethical considerations that demand careful examination. As Safdari et al. [10] emphasize, the increasing integration of LLMs into public-facing applications necessitates a "solid foundation" for responsible AI development, particularly as these models exhibit "synthetic personality embedded in these models" through their training on vast amounts of human data.

One key ethical concern relates to the appropriateness of manipulating the personality traits of LLMs, either through inducing desirable characteristics or suppressing undesirable ones. While our approach focuses on enhancing rather than suppressing traits, it still raises questions about the potential for both beneficial and harmful applications. Inducing seemingly positive traits like helpfulness or patience in a customer service chatbot might improve user experience, but it could also mask the limitations of the underlying technology or create unrealistic expectations of empathy

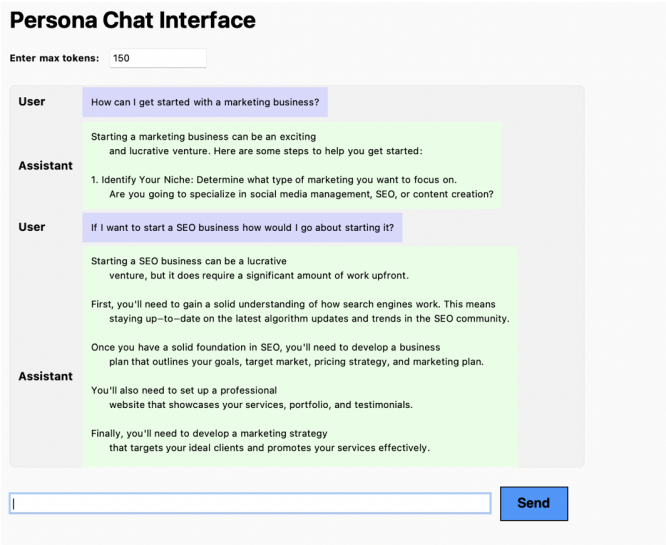

Figure 12: Chat interface for interacting with a custom-designed personality (Friendly and Assertive traits).

or understanding [10]. Furthermore, the dynamic nature of this technology introduces a level of unpredictability. Even with benign traits, the model's responses might deviate from expected behavior, potentially leading to unintended bias or harm.

Our method is consistent with the goal of making AI systems more useful, honest, and harmless. We are aware of the possibility for misuse of AI steering approaches, including our own. For example, it can be used to steer the model toward more damaging, biased, or toxic results. We encourage users of this method to be responsible and avoid increasing harmful behaviors through steering.

6 Future Work

This study advances our understanding of personality traits in large language models (LLMs), but it also offers up new areas for future research. One significant area is to enhance interpretability by mapping personality traits to specific layer activations inside LLMs. Future research should go deeper into these activation patterns, determining which neurons or subnetworks contribute the most significantly to various personality representations. Furthermore, studying how these patterns alter throughout conversations can reveal information about the real-time effects of personality manipulations on language generation. Extending this research to multilingual models will also help us understand the universality versus cultural specificity of personality traits, as well as how cultural factors influence their representation and manifestation across languages.

Another promising direction involves analyzing the performance consequences of different personality factors on LLM competencies. Research reveals that steering vectors, which adjust model